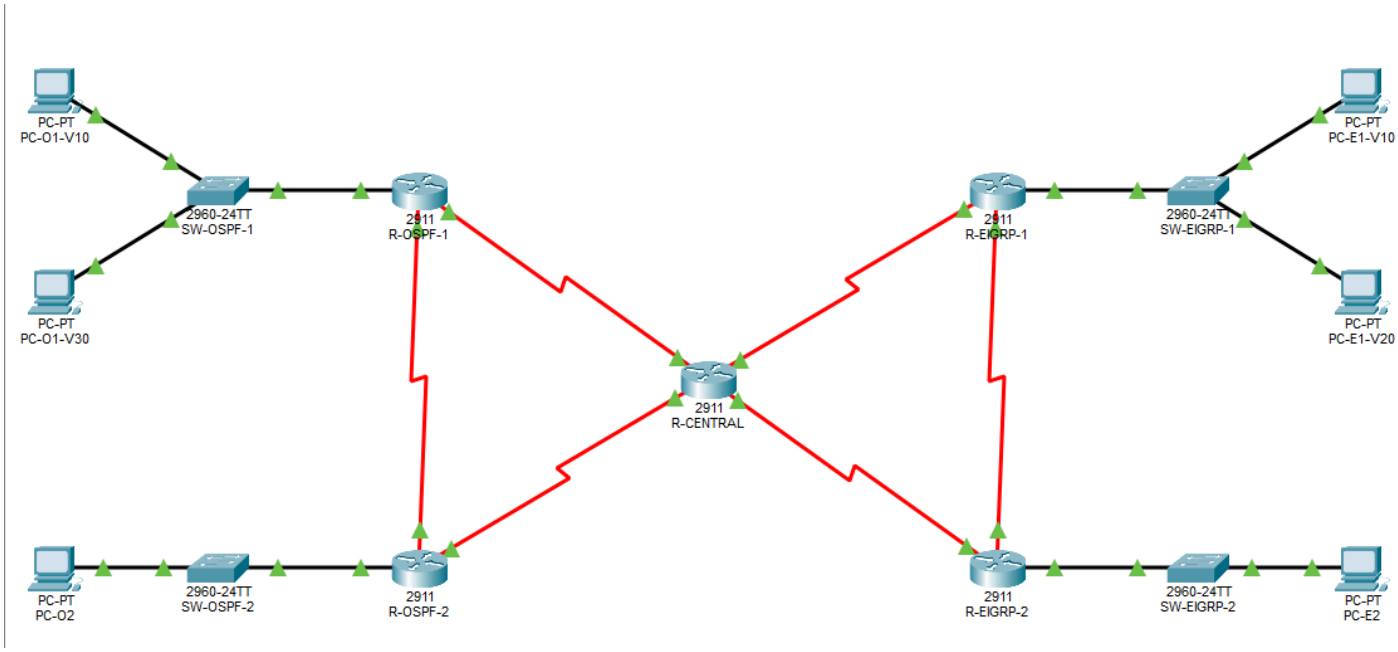
Laboratorio Packet Tracer - PDF2: evidencias de verificacion

Planificacion IP, VLSM, VLANs y enrutamiento OSPF / EIGRP con redistribucion
Curso: Redes - Seccion 10 Docente: Kevin Velasquez
Todas las salidas son capturas reales de la simulacion (archivo entregables/Lab3_Redes.pkt).
Solo se quitaron los restos de paginado (--More--) y las lineas de prompt vacias.

Topologia (captura de pantalla)



1. show ip route de los 5 routers

R-OSPF-1 - show ip route

```
R-OSPF-1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route
Gateway of last resort is not set
     10.0.0.0/8 is variably subnetted, 8 subnets, 2 masks
C       10.0.0.0/30 is directly connected, Serial0/0/0
L       10.0.0.1/32 is directly connected, Serial0/0/0
C       10.0.0.4/30 is directly connected, Serial0/0/1
L       10.0.0.5/32 is directly connected, Serial0/0/1
O       10.0.0.8/30 [110/128] via 10.0.0.6, 00:11:47, Serial0/0/1
O E1    10.0.0.12/30 [110/164] via 10.0.0.6, 00:11:47, Serial0/0/1
O E1    10.0.0.16/30 [110/164] via 10.0.0.6, 00:11:47, Serial0/0/1
O E1    10.0.0.20/30 [110/164] via 10.0.0.6, 00:11:47, Serial0/0/1
     172.16.0.0/16 is variably subnetted, 4 subnets, 3 masks
O E1    172.16.0.0/24 [110/164] via 10.0.0.6, 00:06:14, Serial0/0/1
O E1    172.16.0.0/25 [110/164] via 10.0.0.6, 00:11:47, Serial0/0/1
O E1    172.16.0.128/26 [110/164] via 10.0.0.6, 00:11:47, Serial0/0/1
O E1    172.16.0.192/26 [110/164] via 10.0.0.6, 00:11:47, Serial0/0/1
O E1 192.168.0.0/23 [110/164] via 10.0.0.6, 00:11:37, Serial0/0/1
O    192.168.0.0/24 [110/129] via 10.0.0.2, 00:03:51, Serial0/0/0
                    [110/129] via 10.0.0.6, 00:03:51, Serial0/0/1
     192.168.1.0/24 is variably subnetted, 4 subnets, 3 masks
C       192.168.1.0/26 is directly connected, GigabitEthernet0/0.10
L       192.168.1.1/32 is directly connected, GigabitEthernet0/0.10
C       192.168.1.64/28 is directly connected, GigabitEthernet0/0.30
L       192.168.1.65/32 is directly connected, GigabitEthernet0/0.30
```

R-OSPF-2 - show ip route

```
R-OSPF-2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route
Gateway of last resort is not set
     10.0.0.0/8 is variably subnetted, 8 subnets, 2 masks
C       10.0.0.0/30 is directly connected, Serial0/0/0
L       10.0.0.2/32 is directly connected, Serial0/0/0
O       10.0.0.4/30 [110/128] via 10.0.0.10, 00:03:49, Serial0/0/1
C       10.0.0.8/30 is directly connected, Serial0/0/1
L       10.0.0.9/32 is directly connected, Serial0/0/1
O E1    10.0.0.12/30 [110/164] via 10.0.0.10, 00:03:49, Serial0/0/1
O E1    10.0.0.16/30 [110/164] via 10.0.0.10, 00:03:49, Serial0/0/1
O E1    10.0.0.20/30 [110/164] via 10.0.0.10, 00:03:49, Serial0/0/1
     172.16.0.0/16 is variably subnetted, 4 subnets, 3 masks
O E1    172.16.0.0/24 [110/164] via 10.0.0.10, 00:03:49, Serial0/0/1
O E1    172.16.0.0/25 [110/164] via 10.0.0.10, 00:03:49, Serial0/0/1
O E1    172.16.0.128/26 [110/164] via 10.0.0.10, 00:03:49, Serial0/0/1
O E1    172.16.0.192/26 [110/164] via 10.0.0.10, 00:03:49, Serial0/0/1
O E1 192.168.0.0/23 [110/164] via 10.0.0.10, 00:03:49, Serial0/0/1
     192.168.0.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.0.0/24 is directly connected, GigabitEthernet0/0
L       192.168.0.1/32 is directly connected, GigabitEthernet0/0
     192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
O       192.168.1.0/26 [110/129] via 10.0.0.1, 00:03:49, Serial0/0/0
                       [110/129] via 10.0.0.10, 00:03:49, Serial0/0/1
O       192.168.1.64/28 [110/129] via 10.0.0.1, 00:03:49, Serial0/0/0
                        [110/129] via 10.0.0.10, 00:03:49, Serial0/0/1
```

R-CENTRAL - show ip route

```
R-CENTRAL#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
```

```
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route
Gateway of last resort is not set
     10.0.0.0/8 is variably subnetted, 10 subnets, 2 masks
O       10.0.0.0/30 [110/192] via 10.0.0.5, 00:11:47, Serial0/0/0
                    [110/192] via 10.0.0.9, 00:11:47, Serial0/0/1
D EX    10.0.0.4/30 is directly connected, 00:12:01, Serial0/0/0
L       10.0.0.6/32 is directly connected, Serial0/0/0
D EX    10.0.0.8/30 is directly connected, 00:11:59, Serial0/0/1
L       10.0.0.10/32 is directly connected, Serial0/0/1
D       10.0.0.12/30 [90/2809856] via 10.0.0.17, 00:11:59, Serial0/1/0
                     [90/2809856] via 10.0.0.21, 00:11:57, Serial0/1/1
C       10.0.0.16/30 is directly connected, Serial0/1/0
L       10.0.0.18/32 is directly connected, Serial0/1/0
C       10.0.0.20/30 is directly connected, Serial0/1/1
L       10.0.0.22/32 is directly connected, Serial0/1/1
     172.16.0.0/16 is variably subnetted, 4 subnets, 3 masks
S       172.16.0.0/24 is directly connected, Null0
D       172.16.0.0/25 [90/2172416] via 10.0.0.21, 00:12:00, Serial0/1/1
D       172.16.0.128/26 [90/2172416] via 10.0.0.17, 00:11:59, Serial0/1/0
D       172.16.0.192/26 [90/2172416] via 10.0.0.17, 00:11:59, Serial0/1/0
D    192.168.0.0/23 is a summary, 00:11:47, Null0
O    192.168.0.0/24 [110/65] via 10.0.0.9, 00:11:47, Serial0/0/1
     192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
O       192.168.1.0/26 [110/65] via 10.0.0.5, 00:11:47, Serial0/0/0
O       192.168.1.64/28 [110/65] via 10.0.0.5, 00:11:47, Serial0/0/0
```

R-EIGRP-1 - show ip route

```
R-EIGRP-1>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route
Gateway of last resort is not set
     10.0.0.0/8 is variably subnetted, 8 subnets, 2 masks
D EX    10.0.0.0/30 [170/6752256] via 10.0.0.18, 00:06:19, Serial0/0/1
D EX    10.0.0.4/30 [170/6752256] via 10.0.0.18, 00:11:58, Serial0/0/1
D EX    10.0.0.8/30 [170/6752256] via 10.0.0.18, 00:11:58, Serial0/0/1
C       10.0.0.12/30 is directly connected, Serial0/0/0
L       10.0.0.13/32 is directly connected, Serial0/0/0
C       10.0.0.16/30 is directly connected, Serial0/0/1
L       10.0.0.17/32 is directly connected, Serial0/0/1
D       10.0.0.20/30 [90/7238656] via 10.0.0.18, 00:11:58, Serial0/0/1
                     [90/7366656] via 10.0.0.14, 00:11:57, Serial0/0/0
     172.16.0.0/16 is variably subnetted, 5 subnets, 3 masks
D       172.16.0.0/25 [90/2300416] via 10.0.0.14, 00:11:57, Serial0/0/0
                      [90/7241216] via 10.0.0.18, 00:11:58, Serial0/0/1
C       172.16.0.128/26 is directly connected, GigabitEthernet0/0.10
L       172.16.0.129/32 is directly connected, GigabitEthernet0/0.10
C       172.16.0.192/26 is directly connected, GigabitEthernet0/0.20
L       172.16.0.193/32 is directly connected, GigabitEthernet0/0.20
D    192.168.0.0/23 [90/7238656] via 10.0.0.18, 00:11:47, Serial0/0/1
```

R-EIGRP-2 - show ip route

```
R-EIGRP-2>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route
Gateway of last resort is not set
     10.0.0.0/8 is variably subnetted, 8 subnets, 2 masks
D EX    10.0.0.0/30 [170/6752256] via 10.0.0.22, 00:06:19, Serial0/0/1
D EX    10.0.0.4/30 [170/6752256] via 10.0.0.22, 00:11:59, Serial0/0/1
```

```
D EX    10.0.0.8/30 [170/6752256] via 10.0.0.22, 00:11:59, Serial0/0/1
C       10.0.0.12/30 is directly connected, Serial0/0/0
L       10.0.0.14/32 is directly connected, Serial0/0/0
D       10.0.0.16/30 [90/7238656] via 10.0.0.22, 00:11:59, Serial0/0/1
                     [90/7366656] via 10.0.0.13, 00:11:57, Serial0/0/0
C       10.0.0.20/30 is directly connected, Serial0/0/1
L       10.0.0.21/32 is directly connected, Serial0/0/1
     172.16.0.0/16 is variably subnetted, 4 subnets, 3 masks
C       172.16.0.0/25 is directly connected, GigabitEthernet0/0
L       172.16.0.1/32 is directly connected, GigabitEthernet0/0
D       172.16.0.128/26 [90/2300416] via 10.0.0.13, 00:11:57, Serial0/0/0
                        [90/7241216] via 10.0.0.22, 00:11:59, Serial0/0/1
D       172.16.0.192/26 [90/2300416] via 10.0.0.13, 00:11:57, Serial0/0/0
                        [90/7241216] via 10.0.0.22, 00:11:59, Serial0/0/1
D    192.168.0.0/23 [90/7238656] via 10.0.0.22, 00:11:47, Serial0/0/1
```

2. OSPF: dos rutas de costo igual

Enlace directo R-OSPF-1 <-> R-OSPF-2: Cost 128 (ip ospf cost 128). Via R-CENTRAL: 64 + 64 = 128.
En show ip route ospf, la LAN de R-OSPF-2 aparece con dos caminos [110/129] (10.0.0.2 y 10.0.0.6).

R-OSPF-1 - OSPF vecinos / interfaces / rutas

```
R-OSPF-1#show ip ospf neighbor
Neighbor ID     Pri   State           Dead Time   Address         Interface
5.5.5.5           0   FULL/  -        00:00:37    10.0.0.6        Serial0/0/1
2.2.2.2           0   FULL/  -        00:00:38    10.0.0.2        Serial0/0/0

R-OSPF-1#show ip ospf interface
GigabitEthernet0/0.10 is up, line protocol is up
  Internet address is 192.168.1.1/26, Area 0
  Process ID 1, Router ID 1.1.1.1, Network Type BROADCAST, Cost: 1
  Transmit Delay is 1 sec, State WAITING, Priority 1
  No designated router on this network
  No backup designated router on this network
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    No Hellos (Passive interface)
  Index 1/1, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 0, Adjacent neighbor count is 0
  Suppress hello for 0 neighbor(s)
GigabitEthernet0/0.30 is up, line protocol is up
  Internet address is 192.168.1.65/28, Area 0
  Process ID 1, Router ID 1.1.1.1, Network Type BROADCAST, Cost: 1
  Transmit Delay is 1 sec, State WAITING, Priority 1
  No designated router on this network
  No backup designated router on this network
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    No Hellos (Passive interface)
  Index 2/2, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 0, Adjacent neighbor count is 0
  Suppress hello for 0 neighbor(s)
Serial0/0/0 is up, line protocol is up
  Internet address is 10.0.0.1/30, Area 0
  Process ID 1, Router ID 1.1.1.1, Network Type POINT-TO-POINT, Cost: 128
  Transmit Delay is 1 sec, State POINT-TO-POINT,
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    Hello due in 00:00:00
  Index 3/3, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 1 , Adjacent neighbor count is 1
    Adjacent with neighbor 2.2.2.2
  Suppress hello for 0 neighbor(s)
Serial0/0/1 is up, line protocol is up
  Internet address is 10.0.0.5/30, Area 0
  Process ID 1, Router ID 1.1.1.1, Network Type POINT-TO-POINT, Cost: 64
  Transmit Delay is 1 sec, State POINT-TO-POINT,
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    Hello due in 00:00:00
  Index 4/4, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 1 , Adjacent neighbor count is 1
    Adjacent with neighbor 5.5.5.5
  Suppress hello for 0 neighbor(s)

R-OSPF-1#show ip route ospf
     10.0.0.0/8 is variably subnetted, 8 subnets, 2 masks
O       10.0.0.8 [110/128] via 10.0.0.6, 00:12:10, Serial0/0/1
O E1    10.0.0.12 [110/164] via 10.0.0.6, 00:12:10, Serial0/0/1
O E1    10.0.0.16 [110/164] via 10.0.0.6, 00:12:10, Serial0/0/1
O E1    10.0.0.20 [110/164] via 10.0.0.6, 00:12:10, Serial0/0/1
```

```
     172.16.0.0/16 is variably subnetted, 4 subnets, 3 masks
O E1    172.16.0.0 [110/164] via 10.0.0.6, 00:06:37, Serial0/0/1
O E1    172.16.0.0 [110/164] via 10.0.0.6, 00:12:10, Serial0/0/1
O E1    172.16.0.128 [110/164] via 10.0.0.6, 00:12:10, Serial0/0/1
O E1    172.16.0.192 [110/164] via 10.0.0.6, 00:12:10, Serial0/0/1
     192.168.0.0/24 is variably subnetted, 2 subnets, 2 masks
O E1    192.168.0.0 [110/164] via 10.0.0.6, 00:12:00, Serial0/0/1
O       192.168.0.0 [110/129] via 10.0.0.2, 00:04:14, Serial0/0/0
                    [110/129] via 10.0.0.6, 00:04:14, Serial0/0/1
```

R-OSPF-2 - OSPF vecinos / interfaces / rutas

```
R-OSPF-2#show ip ospf neighbor
Neighbor ID     Pri   State           Dead Time   Address         Interface
1.1.1.1           0   FULL/  -        00:00:30    10.0.0.1        Serial0/0/0
5.5.5.5           0   FULL/  -        00:00:39    10.0.0.10       Serial0/0/1

R-OSPF-2#show ip ospf interface
GigabitEthernet0/0 is up, line protocol is up
  Internet address is 192.168.0.1/24, Area 0
  Process ID 1, Router ID 2.2.2.2, Network Type BROADCAST, Cost: 1
  Transmit Delay is 1 sec, State WAITING, Priority 1
  No designated router on this network
  No backup designated router on this network
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    No Hellos (Passive interface)
  Index 1/1, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 0, Adjacent neighbor count is 0
  Suppress hello for 0 neighbor(s)
Serial0/0/1 is up, line protocol is up
  Internet address is 10.0.0.9/30, Area 0
  Process ID 1, Router ID 2.2.2.2, Network Type POINT-TO-POINT, Cost: 64
  Transmit Delay is 1 sec, State POINT-TO-POINT,
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    Hello due in 00:00:00
  Index 2/2, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 1 , Adjacent neighbor count is 1
    Adjacent with neighbor 5.5.5.5
  Suppress hello for 0 neighbor(s)
Serial0/0/0 is up, line protocol is up
  Internet address is 10.0.0.2/30, Area 0
  Process ID 1, Router ID 2.2.2.2, Network Type POINT-TO-POINT, Cost: 128
  Transmit Delay is 1 sec, State POINT-TO-POINT,
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    Hello due in 00:00:08
  Index 3/3, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 1 , Adjacent neighbor count is 1
    Adjacent with neighbor 1.1.1.1
  Suppress hello for 0 neighbor(s)

R-OSPF-2#show ip route ospf
     10.0.0.0/8 is variably subnetted, 8 subnets, 2 masks
O       10.0.0.4 [110/128] via 10.0.0.10, 00:04:12, Serial0/0/1
O E1    10.0.0.12 [110/164] via 10.0.0.10, 00:04:12, Serial0/0/1
O E1    10.0.0.16 [110/164] via 10.0.0.10, 00:04:12, Serial0/0/1
O E1    10.0.0.20 [110/164] via 10.0.0.10, 00:04:12, Serial0/0/1
     172.16.0.0/16 is variably subnetted, 4 subnets, 3 masks
O E1    172.16.0.0 [110/164] via 10.0.0.10, 00:04:12, Serial0/0/1
O E1    172.16.0.0 [110/164] via 10.0.0.10, 00:04:12, Serial0/0/1
O E1    172.16.0.128 [110/164] via 10.0.0.10, 00:04:12, Serial0/0/1
O E1    172.16.0.192 [110/164] via 10.0.0.10, 00:04:12, Serial0/0/1
     192.168.0.0/24 is variably subnetted, 3 subnets, 3 masks
O E1    192.168.0.0 [110/164] via 10.0.0.10, 00:04:12, Serial0/0/1
     192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
O       192.168.1.0 [110/129] via 10.0.0.1, 00:04:12, Serial0/0/0
                    [110/129] via 10.0.0.10, 00:04:12, Serial0/0/1
```

```
O       192.168.1.64 [110/129] via 10.0.0.1, 00:04:12, Serial0/0/0
                     [110/129] via 10.0.0.10, 00:04:12, Serial0/0/1
```

3. EIGRP: topologia, successor, feasible successor y variance

172.16.0.0/25 en R-EIGRP-1: successor via 10.0.0.14 (FD 2300416, RD 28160) y feasible successor via 10.0.0.18 (7241216/2172416). RD 2172416 < FD 2300416 cumple la condicion de factibilidad; relacion 3.15 con variance 4 => las dos rutas se instalan y el trafico se reparte ~75/25. 'Maximum metric variance 4' en show ip protocols.

R-EIGRP-1 - EIGRP vecinos / topologia / rutas / protocolos

```
R-EIGRP-1#show ip eigrp neighbors
IP-EIGRP neighbors for process 100
H   Address         Interface      Hold Uptime    SRTT   RTO   Q   Seq
                                   (sec)          (ms)        Cnt  Num
0   10.0.0.18       Se0/0/1        11   00:12:50  40     1000  0   98
1   10.0.0.14       Se0/0/0        11   00:12:48  40     1000  0   29

R-EIGRP-1#show ip eigrp topology
IP-EIGRP Topology Table for AS 100/ID(3.3.3.3)
Codes: P - Passive, A - Active, U - Update, Q - Query, R - Reply,
       r - Reply status
P 10.0.0.0/30, 1 successors, FD is 6752256
         via Rstatic (6752256/281600)
P 10.0.0.4/30, 1 successors, FD is 6752256
         via Rstatic (6752256/281600)
P 10.0.0.8/30, 1 successors, FD is 6752256
         via Rstatic (6752256/281600)
P 10.0.0.12/30, 1 successors, FD is 2297856
         via Connected, Serial0/0/0
P 10.0.0.16/30, 1 successors, FD is 6726656
         via Connected, Serial0/0/1
P 10.0.0.20/30, 2 successors, FD is 7238656
         via 10.0.0.18 (7238656/2169856), Serial0/0/1
         via 10.0.0.14 (7366656/6726656), Serial0/0/0
P 172.16.0.0/25, 2 successors, FD is 2300416
         via 10.0.0.14 (2300416/28160), Serial0/0/0
         via 10.0.0.18 (7241216/2172416), Serial0/0/1
P 172.16.0.128/26, 1 successors, FD is 28160
         via Connected, GigabitEthernet0/0.10
P 172.16.0.192/26, 1 successors, FD is 28160
         via Connected, GigabitEthernet0/0.20
P 192.168.0.0/23, 1 successors, FD is 7238656
         via 10.0.0.18 (7238656/2169856), Serial0/0/1

R-EIGRP-1#show ip route eigrp
     10.0.0.0/8 is variably subnetted, 8 subnets, 2 masks
D EX    10.0.0.0/30 [170/6752256] via 10.0.0.18, 00:07:10, Serial0/0/1
D EX    10.0.0.4/30 [170/6752256] via 10.0.0.18, 00:12:49, Serial0/0/1
D EX    10.0.0.8/30 [170/6752256] via 10.0.0.18, 00:12:49, Serial0/0/1
D       10.0.0.20/30 [90/7238656] via 10.0.0.18, 00:12:49, Serial0/0/1
                     [90/7366656] via 10.0.0.14, 00:12:48, Serial0/0/0
     172.16.0.0/16 is variably subnetted, 5 subnets, 3 masks
D       172.16.0.0/25 [90/2300416] via 10.0.0.14, 00:12:48, Serial0/0/0
                      [90/7241216] via 10.0.0.18, 00:12:49, Serial0/0/1
D    192.168.0.0/23 [90/7238656] via 10.0.0.18, 00:12:38, Serial0/0/1

R-EIGRP-1#show ip protocols
Routing Protocol is "eigrp  100 "
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Default networks flagged in outgoing updates
  Default networks accepted from incoming updates
  Redistributing: eigrp 100
  EIGRP-IPv4 Protocol for AS(100)
    Metric weight K1=1, K2=0, K3=1, K4=0, K5=0
    NSF-aware route hold timer is 240
    Router-ID: 3.3.3.3
    Topology : 0 (base)
      Active Timer: 3 min
      Distance: internal 90 external 170
      Maximum path: 4
      Maximum hopcount 100
      Maximum metric variance 4
  Automatic Summarization: disabled
  Automatic address summarization:
  Maximum path: 4
```

```
  Routing for Networks:
     10.0.0.12/30
     10.0.0.16/30
     172.16.0.128/26
     172.16.0.192/26
  Passive Interface(s):
    GigabitEthernet0/0.10
    GigabitEthernet0/0.20
  Routing Information Sources:
    Gateway         Distance      Last Update
    10.0.0.18       90            238351
    10.0.0.14       90            240048
  Distance: internal 90 external 170
```

R-EIGRP-2 - EIGRP vecinos / topologia / rutas / protocolos

```
R-EIGRP-2#show ip eigrp neighbors
IP-EIGRP neighbors for process 100
H   Address         Interface      Hold Uptime    SRTT   RTO   Q   Seq
                                   (sec)          (ms)        Cnt  Num
0   10.0.0.22       Se0/0/1        14   00:12:51  40     1000  0   98
1   10.0.0.13       Se0/0/0        13   00:12:48  40     1000  0   19

R-EIGRP-2#show ip eigrp topology
IP-EIGRP Topology Table for AS 100/ID(4.4.4.4)
Codes: P - Passive, A - Active, U - Update, Q - Query, R - Reply,
       r - Reply status
P 10.0.0.0/30, 1 successors, FD is 6752256
         via Rstatic (6752256/281600)
P 10.0.0.4/30, 1 successors, FD is 6752256
         via Rstatic (6752256/281600)
P 10.0.0.8/30, 1 successors, FD is 6752256
         via Rstatic (6752256/281600)
P 10.0.0.12/30, 1 successors, FD is 2297856
         via Connected, Serial0/0/0
P 10.0.0.16/30, 2 successors, FD is 7238656
         via 10.0.0.22 (7238656/2169856), Serial0/0/1
         via 10.0.0.13 (7366656/6726656), Serial0/0/0
P 10.0.0.20/30, 1 successors, FD is 6726656
         via Connected, Serial0/0/1
P 172.16.0.0/25, 1 successors, FD is 28160
         via Connected, GigabitEthernet0/0
P 172.16.0.128/26, 2 successors, FD is 2300416
         via 10.0.0.13 (2300416/28160), Serial0/0/0
         via 10.0.0.22 (7241216/2172416), Serial0/0/1
P 172.16.0.192/26, 2 successors, FD is 2300416
         via 10.0.0.13 (2300416/28160), Serial0/0/0
         via 10.0.0.22 (7241216/2172416), Serial0/0/1
P 192.168.0.0/23, 1 successors, FD is 7238656
         via 10.0.0.22 (7238656/2169856), Serial0/0/1

R-EIGRP-2#show ip route eigrp
     10.0.0.0/8 is variably subnetted, 8 subnets, 2 masks
D EX    10.0.0.0/30 [170/6752256] via 10.0.0.22, 00:07:10, Serial0/0/1
D EX    10.0.0.4/30 [170/6752256] via 10.0.0.22, 00:12:50, Serial0/0/1
D EX    10.0.0.8/30 [170/6752256] via 10.0.0.22, 00:12:50, Serial0/0/1
D       10.0.0.16/30 [90/7238656] via 10.0.0.22, 00:12:50, Serial0/0/1
                     [90/7366656] via 10.0.0.13, 00:12:48, Serial0/0/0
     172.16.0.0/16 is variably subnetted, 4 subnets, 3 masks
D       172.16.0.128/26 [90/2300416] via 10.0.0.13, 00:12:48, Serial0/0/0
                        [90/7241216] via 10.0.0.22, 00:12:50, Serial0/0/1
D       172.16.0.192/26 [90/2300416] via 10.0.0.13, 00:12:48, Serial0/0/0
                        [90/7241216] via 10.0.0.22, 00:12:50, Serial0/0/1
D    192.168.0.0/23 [90/7238656] via 10.0.0.22, 00:12:38, Serial0/0/1

R-EIGRP-2#show ip protocols
Routing Protocol is "eigrp  100 "
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Default networks flagged in outgoing updates
  Default networks accepted from incoming updates
  Redistributing: eigrp 100
  EIGRP-IPv4 Protocol for AS(100)
    Metric weight K1=1, K2=0, K3=1, K4=0, K5=0
    NSF-aware route hold timer is 240
```

```
  Router-ID: 4.4.4.4
  Topology : 0 (base)
    Active Timer: 3 min
    Distance: internal 90 external 170
    Maximum path: 4
    Maximum hopcount 100
    Maximum metric variance 4
Automatic Summarization: disabled
Automatic address summarization:
Maximum path: 4
Routing for Networks:
   10.0.0.12/30
   10.0.0.20/30
   172.16.0.0/25
Passive Interface(s):
  GigabitEthernet0/0
Routing Information Sources:
  Gateway         Distance      Last Update
  10.0.0.22       90            237091
  10.0.0.13       90            240048
Distance: internal 90 external 170
```

4. R-CENTRAL: vecindades y protocolos (redistribucion mutua)

R-CENTRAL - vecinos y protocolos

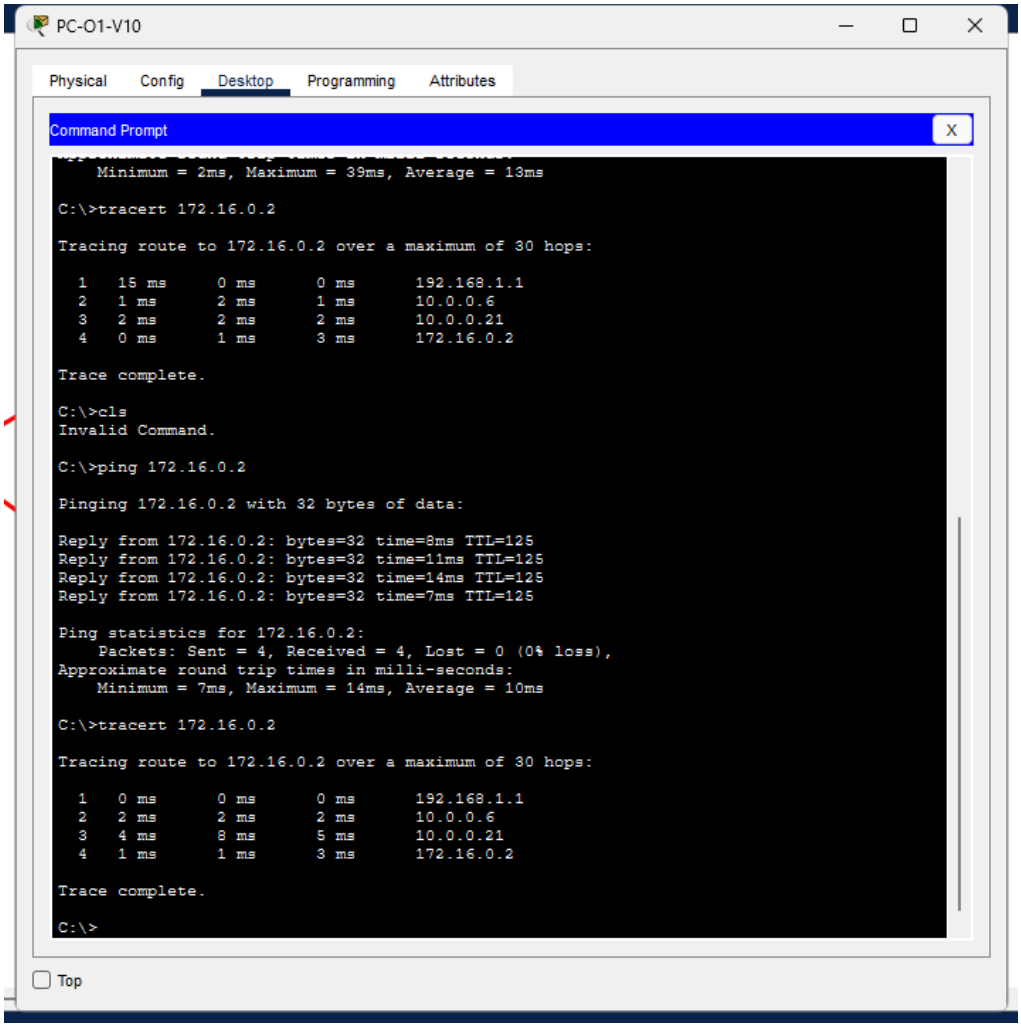
```
R-CENTRAL#show ip ospf neighbor
Neighbor ID     Pri   State           Dead Time   Address         Interface
2.2.2.2           0   FULL/  -        00:00:32    10.0.0.9        Serial0/0/1
1.1.1.1           0   FULL/  -        00:00:32    10.0.0.5        Serial0/0/0

R-CENTRAL#show ip eigrp neighbors
IP-EIGRP neighbors for process 100
H   Address         Interface      Hold Uptime    SRTT   RTO   Q   Seq
                                   (sec)          (ms)        Cnt  Num
0   10.0.0.21       Se0/1/1        14   00:16:01  40     1000  0   30
1   10.0.0.17       Se0/1/0        14   00:16:00  40     1000  0   20

R-CENTRAL#show ip protocols
Routing Protocol is "eigrp  100 "
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Default networks flagged in outgoing updates
  Default networks accepted from incoming updates
  Redistributing: eigrp 100, ospf 1
  EIGRP-IPv4 Protocol for AS(100)
    Metric weight K1=1, K2=0, K3=1, K4=0, K5=0
    NSF-aware route hold timer is 240
    Router-ID: 5.5.5.5
    Topology : 0 (base)
      Active Timer: 3 min
      Distance: internal 90 external 170
      Maximum path: 4
      Maximum hopcount 100
      Maximum metric variance 1
  Automatic Summarization: disabled
  Automatic address summarization:
    192.168.0.0/23 for Serial0/1/0, Serial0/1/1
      Summarizing with metric 2169856
  Maximum path: 4
  Routing for Networks:
     10.0.0.16/30
     10.0.0.20/30
  Routing Information Sources:
    Gateway         Distance      Last Update
    10.0.0.21       90            237090
    10.0.0.17       90            238350
  Distance: internal 90 external 170
Routing Protocol is "ospf 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 5.5.5.5
  It is an autonomous system boundary router
  Redistributing External Routes from,
    eigrp 100
    static
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    10.0.0.4 0.0.0.3 area 0
    10.0.0.8 0.0.0.3 area 0
  Routing Information Sources:
    Gateway         Distance      Last Update
    1.1.1.1              110      00:06:08
    2.2.2.2              110      00:16:02
    5.5.5.5              110      00:06:09
  Distance: (default is 110)
R-CENTRAL con0 is now available
Press RETURN to get started.
```

5. Conectividad extremo a extremo: ping y tracert

Captura de la consola de PC-O1-V10 (OSPF, VLAN 10) hacia PC-E2 (EIGRP, 172.16.0.2):



Transcripciones (ping y tracert entre dominios y VLANs)

PC-O1-V10 -> 172.16.0.2

```
ipconfig
FastEthernet0 Connection:(default port)
   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: FE80::260:2FFF:FE6E:8796
   IPv6 Address....................: ::
   IPv4 Address....................: 192.168.1.2
   Subnet Mask.....................: 255.255.255.192
   Default Gateway.................: ::
                                     192.168.1.1
Bluetooth Connection:
   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: ::
   IPv6 Address....................: ::
   IPv4 Address....................: 0.0.0.0
   Subnet Mask.....................: 0.0.0.0
   Default Gateway.................: ::
                                     0.0.0.0
C:\>ping 172.16.0.2
Pinging 172.16.0.2 with 32 bytes of data:
Reply from 172.16.0.2: bytes=32 time=39ms TTL=125
Reply from 172.16.0.2: bytes=32 time=5ms TTL=125
Reply from 172.16.0.2: bytes=32 time=2ms TTL=125
Reply from 172.16.0.2: bytes=32 time=6ms TTL=125
Ping statistics for 172.16.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 39ms, Average = 13ms
tracert 172.16.0.2
Tracing route to 172.16.0.2 over a maximum of 30 hops:
  1   15 ms     0 ms      0 ms      192.168.1.1
  2   1 ms      2 ms      1 ms      10.0.0.6
  3   2 ms      2 ms      2 ms      10.0.0.21
  4   0 ms      1 ms      3 ms      172.16.0.2
Trace complete.
```

PC-E2 -> 192.168.1.2

```
ipconfig
FastEthernet0 Connection:(default port)
   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: FE80::20C:85FF:FE39:3996
   IPv6 Address....................: ::
   IPv4 Address....................: 172.16.0.2
   Subnet Mask.....................: 255.255.255.128
   Default Gateway.................: ::
                                     172.16.0.1
Bluetooth Connection:
   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: ::
   IPv6 Address....................: ::
   IPv4 Address....................: 0.0.0.0
   Subnet Mask.....................: 0.0.0.0
   Default Gateway.................: ::
                                     0.0.0.0
C:\>ping 192.168.1.2
Pinging 192.168.1.2 with 32 bytes of data:
Reply from 192.168.1.2: bytes=32 time=33ms TTL=125
Reply from 192.168.1.2: bytes=32 time=5ms TTL=125
Reply from 192.168.1.2: bytes=32 time=2ms TTL=125
Reply from 192.168.1.2: bytes=32 time=3ms TTL=125
Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 33ms, Average = 10ms
tracert 192.168.1.2
Tracing route to 192.168.1.2 over a maximum of 30 hops:
  1   13 ms     0 ms      0 ms      172.16.0.1
  2   1 ms      17 ms     1 ms      10.0.0.22
  3   3 ms      1 ms      2 ms      10.0.0.5
  4   1 ms      1 ms      2 ms      192.168.1.2
Trace complete.
```

PC-O1-V30 -> 172.16.0.194

```
ipconfig
FastEthernet0 Connection:(default port)
   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: FE80::2E0:F9FF:FE02:92E
   IPv6 Address....................: ::
   IPv4 Address....................: 192.168.1.66
   Subnet Mask.....................: 255.255.255.240
   Default Gateway.................: ::
                                     192.168.1.65
Bluetooth Connection:
   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: ::
   IPv6 Address....................: ::
   IPv4 Address....................: 0.0.0.0
   Subnet Mask.....................: 0.0.0.0
   Default Gateway.................: ::
                                     0.0.0.0
C:\>ping 172.16.0.194
Pinging 172.16.0.194 with 32 bytes of data:
Reply from 172.16.0.194: bytes=32 time=37ms TTL=125
Reply from 172.16.0.194: bytes=32 time=4ms TTL=125
Reply from 172.16.0.194: bytes=32 time=2ms TTL=125
Reply from 172.16.0.194: bytes=32 time=10ms TTL=125
Ping statistics for 172.16.0.194:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 37ms, Average = 13ms
tracert 172.16.0.194
Tracing route to 172.16.0.194 over a maximum of 30 hops:
  1   9 ms      0 ms      0 ms      192.168.1.65
  2   1 ms      1 ms      1 ms      10.0.0.6
  3   1 ms      1 ms      2 ms      10.0.0.17
  4   0 ms      0 ms      2 ms      172.16.0.194
Trace complete.
```

PC-E1-V20 -> 192.168.1.66

```
ipconfig
FastEthernet0 Connection:(default port)
   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: FE80::2D0:BCFF:FE99:3597
   IPv6 Address....................: ::
   IPv4 Address....................: 172.16.0.194
   Subnet Mask.....................: 255.255.255.192
   Default Gateway.................: ::
                                     172.16.0.193
Bluetooth Connection:
   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: ::
   IPv6 Address....................: ::
   IPv4 Address....................: 0.0.0.0
   Subnet Mask.....................: 0.0.0.0
   Default Gateway.................: ::
                                     0.0.0.0
C:\>ping 192.168.1.66
Pinging 192.168.1.66 with 32 bytes of data:
Reply from 192.168.1.66: bytes=32 time=27ms TTL=125
Reply from 192.168.1.66: bytes=32 time=5ms TTL=125
Reply from 192.168.1.66: bytes=32 time=5ms TTL=125
Reply from 192.168.1.66: bytes=32 time=4ms TTL=125
Ping statistics for 192.168.1.66:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 4ms, Maximum = 27ms, Average = 10ms
tracert 192.168.1.66
Tracing route to 192.168.1.66 over a maximum of 30 hops:
  1   6 ms      4294967295 ms0 ms      172.16.0.193
  2   1 ms      1 ms      1 ms      10.0.0.18
  3   2 ms      1 ms      3 ms      10.0.0.5
  4   0 ms      0 ms      4 ms      192.168.1.66
Trace complete.
```

PC-O2 -> 172.16.0.130

```
ipconfig
FastEthernet0 Connection:(default port)
   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: FE80::2E0:8FFF:FE22:BC24
   IPv6 Address....................: ::
   IPv4 Address....................: 192.168.0.2
   Subnet Mask.....................: 255.255.255.0
   Default Gateway.................: ::
                                     192.168.0.1
Bluetooth Connection:
   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: ::
   IPv6 Address....................: ::
   IPv4 Address....................: 0.0.0.0
   Subnet Mask.....................: 0.0.0.0
   Default Gateway.................: ::
                                     0.0.0.0
C:\>ping 172.16.0.130
Pinging 172.16.0.130 with 32 bytes of data:
Reply from 172.16.0.130: bytes=32 time=11ms TTL=125
Reply from 172.16.0.130: bytes=32 time=3ms TTL=125
Reply from 172.16.0.130: bytes=32 time=3ms TTL=125
Reply from 172.16.0.130: bytes=32 time=7ms TTL=125
Ping statistics for 172.16.0.130:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 3ms, Maximum = 11ms, Average = 6ms
tracert 172.16.0.130
Tracing route to 172.16.0.130 over a maximum of 30 hops:
  1   3 ms      1 ms      0 ms      192.168.0.1
  2   1 ms      1 ms      2 ms      10.0.0.10
  3   3 ms      2 ms      2 ms      10.0.0.17
  4   32 ms     2 ms      1 ms      172.16.0.130
Trace complete.
```

PC-E1-V10 -> 192.168.0.2

```
ipconfig
FastEthernet0 Connection:(default port)
   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: FE80::290:CFF:FE1E:E6A3
   IPv6 Address....................: ::
   IPv4 Address....................: 172.16.0.130
   Subnet Mask.....................: 255.255.255.192
   Default Gateway.................: ::
                                     172.16.0.129
Bluetooth Connection:
   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: ::
   IPv6 Address....................: ::
   IPv4 Address....................: 0.0.0.0
   Subnet Mask.....................: 0.0.0.0
   Default Gateway.................: ::
                                     0.0.0.0
C:\>ping 192.168.0.2
Pinging 192.168.0.2 with 32 bytes of data:
Reply from 192.168.0.2: bytes=32 time=13ms TTL=125
Reply from 192.168.0.2: bytes=32 time=4ms TTL=125
Reply from 192.168.0.2: bytes=32 time=2ms TTL=125
Reply from 192.168.0.2: bytes=32 time=3ms TTL=125
Ping statistics for 192.168.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 13ms, Average = 5ms
tracert 192.168.0.2
Tracing route to 192.168.0.2 over a maximum of 30 hops:
  1   0 ms      0 ms      0 ms      172.16.0.129
  2   2 ms      1 ms      1 ms      10.0.0.18
  3   1 ms      1 ms      4 ms      10.0.0.9
  4   18 ms     4 ms      1 ms      192.168.0.2
Trace complete.
```